Diagnosis Functional Description

ABS/ESP-System

RoutineControl Section

Protocol: ***UDS*** (ISO 14229-1)
Communication Bus: ***CAN***

1.1.1 Service $31 – RoutineControl

1.1.1.1. Description

The Routine Control service is used by a diagnostic tool to start a routine, to stop a routine and to request the routine results.

Note that the actual description of the end-of-line testing, evacuation/filling and repair bleeding sequences are described in separate documents, provided by CC/ESU1.

1.1.1.2. Availability

Supported Diagnostic Sessions		defaultSession
	X	extendedDiagnosticSession
Supported Addressing Modes	**X**	physical
		functional

1.1.1.3. Request Message

Refer to the sub-sections for each routine control type and identifier for details.

1.1.1.4. Response Message

Refer to the sub-sections for each routine control type and identifier for details.

1.1.1.5. Negative Response Codes

Value	Description
$12	Sub Function Not Supported
$13	Incorrect Message Length
$22	Conditions Not Correct
$24	Request Sequence Error
$31	Request Out Of Range
$78	Request Correctly Received – Response Pending

1.1.1.6. Routine Control Type $01 – Start Routine

1.1.1.6.1. Response Message

Byte	Bit	Description	Value + Unit	LSB	Offset
1		Response Service Id = [Routine Control]	$71		
2		Sub Function = [Start Routine]	$01		
3..4		Routine Identifier	$xxxx		
5		Routine Status Byte	$01 = start successful $02 = already started $03 = start not possible *Only used if no RequestRoutineResult identifier is supported:*		

			$04 = finished $05 = routine error		

1.1.1.6.2. Identifier $F000 – EvacAndFill

Byte	Bit	Description	Coding	Hex Val
1		Request Service Id = [Routine Control]		31
2		Sub Function Routine Control Type [01h] = Start Routine		01
3 4		Routine Identifier = [Evac and Fill] Byte #1 Routine Identifier High Byte Byte #2 Routine Identifier Low Byte		F0 00
5		Evac and Fill Time Format: 1 Bit = 1s (unsigned Hex) Range: 1s … 150s (01h … 96h)	Hex	01-96
6		Time till Start of Pump Motor Resolution: 1 Bit = 1s (unsigned Hex) Range:: [01h … FEh] = 1 … 254s [00h] = 1s [FFh] = "Pump motor is not running"	Hex	00-FF

Evac and fill time

After evac and filling time, the pump motor will run for 2 additional seconds, if the delay time in byte 6 is not $FF.

1.1.1.6.3. Identifier $F001 – RepairBleed

Byte	Bit	Description	Coding	Hex Val
1		Request Service Id = [Routine Control]		31
2		Sub Function Routine Control Type [01h] = Start Routine		01
3 4		Routine Identifier = [Repair Bleed] Byte #1 Routine Identifier High Byte Byte #2 Routine Identifier Low Byte	 Hex Hex	F0 01
5		Repair Bleeding Phase Identifier [01h] = EOL Repair Bleeding Phase 1 [02h] = EOL Repair Bleeding Phase 2 [03h] = EOL Repair Bleeding Phase 3 [04h] = EOL Repair Bleeding Phase 4	Hex	01-04
6		Number middle loop repetitions [03h] = 3 repetitions [04h] = 4 repetitions [05h] = 5 repetitions This Parameter is always required, but only valid for Phase 3, 4	Hex	03-05

Repair bleeding phase

The Repair Bleeding function should always be executed in increasing phase sequence starting with Phase 1. A repetition of the same phase is allowed, but no phase shall be skipped. If this is not assured, the pump might be damaged due to dry run.

Repetition counter

This parameter defines the number of inner loop repetitions in phase 3 and 4. This Parameter is always required, but only valid for Phase 3, 4. For other phases, it is ignored.

1.1.1.6.4. Identifier $F002 – WheelSpeedSensorTest

The “Wheel Speed Sensor Test” is used to check the correct connection, correlation and function of the wheel speed sensors as well as a damaged multi pole wheel. A wrong air gap or eccentricity will also be detected. The test can be executed for both acceleration and deceleration.
The “Wheel Speed Sensor Test” requires a brake dynamometer with the possibility to actuate each wheel on the test bench.

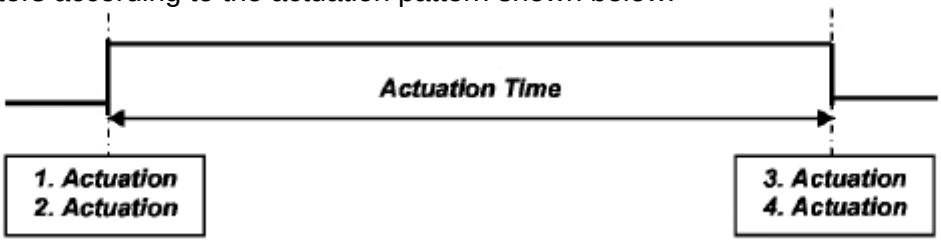
Byte	Bit	Description	Coding	Hex Val
1		Request Service Id = [Routine Control]		31
2		Sub Function Routine Control Type [01h] = Start Routine		01
3 4		Routine Identifier = [Wheel Speed Sensor Test] Byte #1 Routine Identifier High Byte Byte #2 Routine Identifier Low Byte	 Hex Hex	F0 02
5-6		V_{Stopp} Format 1 Bit = 0,01km/h (Unsigned Hex) Range: 0km/h ... 655,35km/h (theoretically)[0000h ... FFFFh]	 Hex Hex	 00-FF 00-FF
7		Testing Phase Format [00h] = Deceleration; [01h] = Acceleration Reserved (set to zero)	Hex	00-01

1.1.1.6.5. Identifier $F003 - StaticTest

Byte	Bit	Description	Value + Unit	LSB	Offset
1		Request Service Id = [Routine Control]	$31		
2	0..6	Sub Function = [Start Routine]	$01		
	7	Suppress Positive Response Bit	0 = ECU is sending a positive response 1 = ECU is not sending a positive response		
3..4		Routine Identifier = [StaticTest]	$F003		

1.1.1.6.6. Identifier $F004 – Actuator Test

The Actuator Test will be used to check the hydraulic connections, the tightness and correlation ofthe valves, the function of the pump and the air in the brake system. This test controls up to 4 actuators according to the actuation pattern shown below.



Byte	Bit	Description	Coding	Hex Val
1		Request Service Id = [Routine Control]		31
2		Sub Function Routine Control Type [01h] = Start Routine		01
		Routine Identifier = [Actuator Test]		
3		Byte #1 Routine Identifier High Byte	Hex	F0
4		Byte #2 Routine Identifier Low Byte	Hex	04
		Actuation 1		
5		Byte #1 Actuator 1 Value High Byte	Hex	00-00
6		Byte #2 Actuator 1 Value Low Byte	Hex	00-64
7		Byte #3 Actuator 1 Identifier High Byte	Hex	00-FF
8		Byte #4 Actuator 1 Identifier Low Byte	Hex	00-FF
		Actuation 2		
9		Byte #1 Actuator 2 Value High Byte	Hex	00-00
10		Byte #2 Actuator 2 Value Low Byte	Hex	00-64
11		Byte #3 Actuator 2 Identifier High Byte	Hex	00-FF
12		Byte #4 Actuator 2 Identifier Low Byte	Hex	00-FF
		Actuation Time		
13		Byte #1 Actuation Time High Byte	Hex	00-FF
14		Byte #2 Actuation Time Low Byte	Hex	00-FF
		Format 1 Bit = 10 ms (Unsigned Hex)		
		Range: 10 ms ... 327,67 s [0001h ... 7FFFh] other values are undefined!		
		Actuation 3		
15		Byte #1 Actuator 3 Value High Byte	Hex	00-00
16		Byte #2 Actuator 3 Value Low Byte	Hex	00-64
17		Byte #3 Actuator 3 Identifier High Byte	Hex	00-FF
18		Byte #4 Actuator 3 Identifier Low Byte	Hex	00-FF
		Actuation 4		
19		Byte #1 Actuator 4 Value High Byte	Hex	00-00
20		Byte #2 Actuator 4 Value Low Byte	Hex	00-64
21		Byte #3 Actuator 4 Identifier High Byte	Hex	00-FF
22		Byte #4 Actuator 4 Identifier Low Byte	Hex	00-FF

1.1.1.6.7. Identifier $F100 - SteeringAngleSensorCalibration

This Identifier is not supported for ABS

Byte	Bit	Description	Value + Unit	LSB	Offset
1		Request Service Id = [Routine Control]	$31		
2	0..6	Sub Function = [Start Routine]	$01		
	7	Suppress Positive Response Bit	0 = ECU is sending a		

Byte	Bit	Description	Value + Unit	LSB	Offset
			positive response 1 = ECU is not sending a positive response		
3..4		Routine Identifier = [SASCalibration]	$F100		

1.1.1.7. Routine Control Type $02 - Stop Routine

1.1.1.7.1. Request Message

Byte	Bit	Description	Value + Unit	LSB	Offset
1		Request Service Id = [Routine Control]	$31		
2	0..6	Sub Function = [Start Routine]	$02		
	7	Suppress Positive Response Bit	0 = ECU is sending a positive response 1 = ECU is not sending a positive response		
3..4		Routine Identifier	$F000 = EvacFill $F001 = RepairBleed $F002 = WSSTest $F003 = StaticTest $F004=ActuatorTest		

1.1.1.7.2. Response Message

Byte	Bit	Description	Value + Unit	LSB	Offset
1		Response Service Id = [Routine Control]	$71		
2		Sub Function = [Start Routine]	$02		
3..4		Routine Identifier	$xxxx		
5		Routine Status Byte	$01 = stop successful $02 = already stopped or not started $03 = routine stopped abnormally		

1.1.1.8. Routine Control Type $03 - Request Routine Results

1.1.1.8.1. Request Message

Byte	Bit	Description	Value + Unit	LSB	Offset
1		Request Service Id = [Routine Control]	$31		
2	0..6	Sub Function = [RoutineResult]	$03		
	7	Suppress Positive Response Bit	0 = ECU is sending a positive response 1 = ECU is not sending a positive response		
3..4		Routine Identifier	$xxxx		

1.1.1.8.2. Identifier $F000 - EvacAndFill

Byte	Bit	Description	Value + Unit	LSB	Offset
1		Response Service Id = [Routine Control]	$71		
2		Sub Function = [RoutineResult]	$03		

3..4		Routine Identifier = [EvacAndFill]	$F000		
5		Routine Status Byte	01 = Routine completed successfully 02 = Routine not yet completed 03 = Routine canceled by ECU 04 = Routine Identifier not correct 05 = Routine cancelled by Tester 06 = Routine was not started		

1.1.1.8.3. Identifier $F001 - RepairBleed

Byte	Bit	Description	Value + Unit	LSB	Offset
1		Response Service Id = [Routine Control]	$71		
2		Sub Function = [RoutineResult]	$03		
3..4		Routine Identifier = [RepairBleed]	$F001		
5		Routine Status Byte	01 = Routine completed successfully 02 = Routine not yet completed 03 = Routine canceled by ECU 04 = Routine Identifier not correct 05 = Routine cancelled by Tester 06 = Routine was not started		

1.1.1.8.4. Identifier $F002 - WheelSpeedSensorTest

Byte	Bit	Description	Value + Unit
1		Response Service Id = [Routine Control]	$71
2		Sub Function = [RoutineResult]	$03
3..4		Routine Identifier = [WheelSpeedSensorTest]	$F002
5		Routine Status Byte	01 = Routine completed successfully 02 = Routine not yet completed 03 = Routine canceled by ECU 04 = Routine Identifier not correct 05 = Routine cancelled by Tester 06 = Routine was not started
6 .. 7		Vst FL Byte #1 High Byte Byte #2 Low Byte Format 1 Bit = 0,01km/h (Unsigned Hex) Range: 0km/s ... 655,35km/s (theoretically)[0000h ... FFFFh]	00-FF 00-FF
8 .. 9		Vst FR Byte #1 High Byte Byte #2 Low Byte Format 1 Bit = 0,01km/h (Unsigned Hex) Range: 0km/s ... 655,35km/s (theoretically)[0000h ... FFFFh]	00-FF 00-FF
10 .. 11		Vst RL Byte #1 High Byte Byte #2 Low Byte Format 1 Bit = 0,01km/h (Unsigned Hex) Range: 0km/s ... 655,35km/s (theoretically)[0000h ... FFFFh]	00-FF 00-FF
12 .. 13		Vst RR Byte #1 High Byte Byte #2 Low Byte Format 1 Bit = 0,01km/h (Unsigned Hex) Range: 0km/s ... 655,35km/s (theoretically)[0000h ... FFFFh]	00-FF 00-FF
14 .. 15		dV FL Byte #1 High Byte Byte #2 Low Byte Format 1 Bit = 0,01km/h (Unsigned Hex) Range: 0km/s ... 655,35km/s (theoretically)[0000h ... FFFFh]	00-FF 00-FF
16 .. 17		dV FR Byte #1 High Byte Byte #2 Low Byte Format 1 Bit = 0,01km/h (Unsigned Hex) Range: 0km/s ... 655,35km/s (theoretically)[0000h ... FFFFh]	00-FF 00-FF
18 .. 19		dV RL Byte #1 High Byte Byte #2 Low Byte Format 1 Bit = 0,01km/h (Unsigned Hex) Range: 0km/s ... 655,35km/s (theoretically)[0000h ... FFFFh]	00-FF 00-FF
20 .. 21		dV RR Byte #1 High Byte Byte #2 Low Byte Format 1 Bit = 0,01km/h (Unsigned Hex)	00-FF 00-FF

		Range: 0km/s ... 655,35km/s (theoretically)[0000h ... FFFFh]	

1.1.1.8.5. Identifier $F003 - StaticTest

Byte	Bit	Description	Value + Unit	LSB	Offset
1		Response Service Id = [Routine Control]	$71		
2		Sub Function = [RoutineResult]	$03		
3..4		Routine Identifier = [DynamicTest]	$F003		
5		Routine Status Byte	01 = Routine completed successfully 02 = Routine not yet completed 03 = Routine canceled by ECU 04 = Routine Identifier not correct 05 = Routine cancelled by Tester 06 = Routine was not started		

1.1.1.8.6. Identifier $F004 – ActuatorTest

Byte	Bit	Description	Value + Unit	LSB	Offset
1		Response Service Id = [Routine Control]	$71		
2		Sub Function = [RoutineResult]	$03		
3..4		Routine Identifier = [ActuatorTest]	$F004		
5		Routine Status Byte	01 = Routine completed successfully 02 = Routine not yet completed 03 = Routine canceled by ECU 04 = Routine Identifier not correct 05 = Routine cancelled by Tester 06 = Routine was not started		

1.2. Application: End-Of-Line Routines

Note that the actual description of the end-of-line testing, evacuation/filling and repair bleeding sequences are described in separate documents, provided by CC/ESU1.

1.2.1. Actuator Table

The following table lists the available actuator identifiers for service $31..

Actuator – ID	Channel	Actuator – Value
$0000	No activation	
$0022	Pump-Motor (RFP)	[$0064] = on [$0000] = off
$0030	EV front left (EVVL) (Inlet Valve FL)	[$0064] = on [$0000] = off
$0032	AV front left (AVVL) (Outlet Valve FL)	[$0064] = on [$0000] = off
$0034	EV front right (EVVR) (Inlet Valve FR)	[$0064] = on [$0000] = off
$0036	AV front right (AVVR) (Outlet Valve FR)	[$0064] = on [$0000] = off
$0038	EV rear right (EVHR) (Inlet Valve RR)	[$0064] = on [$0000] = off
$003A	AV rear right (AVHR) (Outlet Valve RR)	[$0064] = on [$0000] = off
$003C	EV rear left (EVHL) (Inlet Valve RL)	[$0064] = on [$0000] = off
$003E	AV rear left (AVHL) (Outlet Valve RL)	[$0064] = on [$0000] = off
$004E	Control Valve USV 1	[$0064] = on [$0000] = off
$0050	Control Valve USV 2	[$0064] = on [$0000] = off
$0052	High Pressure Control Valve (HSV) 1	[$0064] = on [$0000] = off
$0054	High Pressure Control Valve (HSV) 2	[$0064] = on [$0000] = off
$0062	Disable / Enable Overheat Protection	[$00A5] = Disable [$0000] = Enable
$0064	Disable / Enable (Value Relais) VR-Shut down	[$00AB] = Disable [$0000] = Enable

1.2.2. Actuation Limitations

Valves and pump motor are covered by protection measures to avoid destruction due to excessive actuation. In case the protection measures are triggered, a fault entry is stored in the fault memory to document the overheat situation.
Protections measures can be disabled by usage of actuator 0062. Actuation of valves or pump with disabled protection measures may damage the brake system due to excess temperature or excess pressure conditions.